EXP NO : 9

Implementation of Subnetting in Cisco Packet Tracer

23/9/24

AIM

To implement class IP subnetting and configure network devices using Cisco Packet Tracer.

PROCEDURE

1) Create a new network topology in Cisco Packet Tracer

2) Add routers, switches, and PC's, then connect them using suitable cables.

3) Divide the network 192.168.1.0/24 into smaller subnet using a /27 subnet mask

4) Configure router interface with IP address and enable them using no shutdown

5) Configure switch ports in access mode

6) Assign static IP addresses, subnet masks, and default gateways to all PCs.

IP Address Configuration

Network	Router IP	PC IP Addresses
192.168.1.0/27	192.168.1.1	.11 - .15
192.168.2.0/27	192.178.1.2	.11 - .14
192.178.3.0/27	192.181.12	.10 - .14

Subnet mask: 255.255.255.224

Router config
enable
configure
interface Fa
ip address
no shut do
exit

Testing
1. Open
2. Use
3. Verify
successfull

RESULT:
Thus
successfu
connec
comm

Router configuration commands

enable
configure terminal
interface FastEthernet 0/0
ip address 192.168.1.1 255.255.255.224
no shutdown
exit

Testing

1. Open the command prompt on each PC.
2. Use the ping command to test connectivity
3. Verify that the devices can communicate successfully.

RESULT:

Thus, classless IP subnetting was implemented successfully using Cisco Packet tracer, and network connectivity was tested using the ping command.